

EXPLORATORY DATA ANALYSIS OF LARGE FOOTBALL DATA

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Exploratory Data Analysis of Large Football Data

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Football data science is a rapidly growing field. Most of the world's top clubs are utilizing it to gain advantage in all aspects of the game. In this project, I explore how two multivariate statistical methods can be applied to this field, specifically k-means clustering and t-distributed stochastic neighbor embedding. Firstly, I set up the notation we will use throughout and give a brief explanation of the methods I will use. Statsbomb (a football data analysis company) have published free event data from the 2015/2016 Premier League which will be the primary data used for this project. I apply k-means clustering and t-SNE to three player summary datasets which are derived from the Statsbomb events data. Each of these summarizes the player's performance according to different areas of the game. I analyze the clusters, and use t-SNE to visualize the data. Finally, I suggest possible applications where this could be useful in football.

Declaration

I declare this thesis is my original work. No portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Chapter 1

Introduction

Data Science is a scientific field of research that dates back to the 18th century, where the field of statistics was first developed. Early statisticians such as Karl Pearson laid the foundations for statistical analysis, data collection, and statistical inference (Norton, 1978). The potential to use computers in the field of statistics was seen and taken advantage of very early after the first computers were created. During the 1960's and 1970's, data warehousing systems were developed which could store and manage large volumes of data (Vaisman and Zimányi, 2014). From this point the field has seen steady advancements, such as better database technology and data visualization methods. The early 21st century the rise of machine learning and artificial intelligence. Here, algorithms were developed which feed computers data, allowing computers to learn and develop the ability to predict future outcomes. This is a field which is developing exponentially (Zhou, 2021). Advancements in the internet and social media have meant the volume, variety, and velocity of data has increased phenomenally . Super-large data sets are frequently available now, known as 'Big Data'. Current advancements in data science are centered around working with Big Data.

Over the last 60 years, the progress in data science has had an impact in multiple industries. Business owners have had an attitude shift towards making decisions that are backed up by data. This has led to evolution in banking, finance, marketing, healthcare, and more recently, *sport*.

A key example where data science first had a big impact in sport was the Oakland A's 2002 season in Major League Baseball. The events of this season are described

in Michael Lewis' #1 times' best seller book "MoneyBall" (Lewis, 2003). This book describes how the Oakland A's manager, Billy Beane, looked outside of classical scouting methods and focused instead purely on statistics when deciding which players to sign. For example, Beane assigned high value to players with a high "on base percentage" (Wasserman et al., 2005, p. 2). This approach led to the A's achieving an unprecedented 20 consecutive wins, outperforming teams with vastly higher budgets such as the Boston Red Sox.

This success was noticed all over the world, in particular, Tottenham Hotspur's director of football, Damien Commoli, came across it in 2005 when he read Moneyball (Graham, 2024, p. 22). He felt he could use data analysis in a similar way to achieve success in football. This led him to appointing Ian Graham, a data scientist who was working with *The Times* to produce football analysis, as head of data science at Tottenham (Graham, 2024, p. 21). Graham was one of the key figures in early football data analysis. He created some early expected goals and possession value models, which are used heavily in football data science today.

Expected Goals is a model that calculates the probability of scoring a goal from any given shot. To do this, we can use the following pieces of information that can be recorded for all shots (Analyst, 2023):

1. Distance from the goal
2. Angle to the goal
3. Pressure the shooter is under from defenders
4. Pattern of play (open play, free kick, corner, etc)
5. Type of assist (cross, throughball, etc)
6. Goalkeeper position

A model can then use this information to predict the expected goals (or $E[g]$) of the shot. One way this could be done is to use some form of regression model:

$$E[g] = a_0 + a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6$$

where the x_i are predictor variables corresponding to the pieces of information listed above, and the a_i are the coefficients that tell you how much each predictor variable affects the expected goals.

Later, Graham went on to work for Liverpool FC. He played a key role in securing Liverpool's first Premier League title in 30 years, with his models being used to help Liverpool sign key players such as Virgil Van Dijk and Mohamed Salah (Graham, 2024, p. 11).

Football data analysis is still a very young and growing field. Manchester City and Liverpool currently lead the way for clubs in the Premier League, both having teams of 10–20 data scientists. Brighton and Brentford are examples of clubs who have used data to their advantage, gaining promotion from the lower tiers of English football into the Premier League. They are both now well established Premier League clubs, and have achieved this with much lower budgets than the large clubs with bigger histories.

The growth of the field has also meant a vast increase in the quantity of data collected. Today there are three main types of Football data collected (Gudmundsson et al., 2017)[p 22:5]:

1. Event Data
2. Tracking Data
3. Advanced Tracking Data

Event data records all the events in a match that the ball is involved with (e.g passes, shots, tackles, dribbles). There are typically around 3400 of these in each match (Statsbomb, 2025). Tracking data records the position of all players on the pitch 25 times every second. This results in over one million tracking locations recorded for every game (Statsbomb, 2025). The advanced tracking data records locations of players limbs and head also, an extension of the tracking data. 360 data combines event data with tracking data to give a full description of any given match (Statsbomb, 2025).

Opta (a sport statistics company) were the trailblazers for collecting football data like this, and its their event data that Graham used to build his models (Graham, 2024, p. 22). Many more data collectors have followed in Opta's footsteps meaning

there is now immense amounts of football data available. Statsbomb (2025) are one of the leading data collectors in football today. Statsbomb are a company solely focused on football data analysis. They collect data, and work with clubs such as Porto and Club Brugge. They give these clubs access to all the data they collect, and help them use it to create models for player recruitment and tactics. Statsbomb also has an aim to grow the field of football data. As a result, they release some of the event data they collect to the public for free. This allows students, or anyone looking to start a career in football data, to use their data for their own research. We use Statsbomb's free event data for this project.

The event data comes in the form of a large table, where each row corresponds to a unique event. We focus on the 2015/2016 Premier League season in this project, and we have access to the event data collected from all matches of this season. This is a total of 380 matches, meaning the events table we load has over a million unique events. While the data is in this form, it is not easy to work with, so much of our work requires transforming the data into more convenient forms.

In this project, I will apply k-means clustering to various player summary tables derived from the event data, and use t-SNE plots as an accompanying visual representation of the data. Most of the popular football data science techniques currently implemented use regression to model different aspects of the game. In this paper, we instead look to see what results can be obtained by applying clustering methods as opposed to regression. In this sense, this project is a form of exploratory data science. I chose k-means as it is a simple, non-model-based form of clustering. This will allow us to see if basic clustering ideas can be applied to football data in a way that provides value.

In particular, I use these methods to compare player performance throughout the season. I hope this will help us identify the top performers for the season, and which metrics separate those that excel from the rest. I look to find players who performed well according to our analysis that perhaps went under the radar in the general media analysis of the season.

I transform the raw data set in different ways to analyze various aspects of the game. The first aspect of the game I wish to explore is finishing ability of attackers. For this, we transform the data so it only contains finishing metrics such as number

of shots and number of shots on target for each player. We only consider players that were strikers or wingers here, as finishing metrics are most relevant for these players. Secondly, we wish to explore passing ability of midfielders. We perform a separate transformation in which we record a number of key passing metrics about the players. For this purpose we only consider midfielders, as passing statistics are most relevant to these players. We wish to see how the passing metrics differ depending on the different types of midfielder. We look to answer questions such as, do attacking midfielders make fewer passes than defensive midfielders? What other passing metrics differ between attacking midfielders and defensive midfielders? Finally, we look to explore the creative ability of attacking midfielders. We transform the data into a form where it contains attacking creativity metrics. We consider attacking and wide midfielders, as well as wingers for this purpose.

Chapter 2

Methods

In this section, we introduce and give some background on the various methods we use throughout this project.

2.1 Notation

Firstly, we introduce and define some common terms in multivariate statistics that we will use throughout this report. For reference to the basic definitions, see the introductory statistics textbook (Ross, 2017).

2.1.1 Random variables

A random variable in statistics is a function which assigns a real value to an outcome in the sample space of an experiment. By sample space we mean all possible outcomes of the experiment. Random variables can be classified into two different types:

1. Discrete random variables
2. Continuous random variables

Discrete random variables can take a finite amount of values. For example, shoe size is a discrete random variable, as there is a finite set of possible shoe sizes. Continuous random variables can take an infinite number of values in a given range. Height is an example of a continuous random variable as it can take any value (in a reasonable range, say 4–8 feet for most adults).

An important property of continuous random variables is their probability density function, or pdf. A pdf is a function which is used to calculate the probability that a random observation will take value in some given range. This is a key property that defines the different types of continuous distributions, and later in this section we show two examples, the normal distribution and Student-t distribution.

Random variables can be univariate or multivariate. Univariate means we are measuring a single variable. Multivariate random variables measure the outcomes of multiple experiments all corresponding to the same observation. In general, multivariate random variables are denoted in vector form e.g.

$$\mathbf{x}_i^T = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{im} \end{bmatrix} \quad (2.1)$$

The vector $\mathbf{x}_i^T$ above represents a multivariate random variable with m variables. Since multivariate random variables are vectors, we use bold face to highlight that we are referring to a vector. We refer to the m individual variables as the 'features' of $\mathbf{x}_i$. We often work with a collection of n observations of a multivariate random variable e.g.

$$\mathbf{x}_1^T, \mathbf{x}_2^T, \dots, \mathbf{x}_n^T$$

represents a collection of n observations, and each observation $\mathbf{x}_i^T$ is a multivariate random variable vector with m features. We will refer to individuals observations as 'samples' from a collection. In many statistical models it is useful when each samples' outcome is independent to the other samples, and they are all identically distributed. We call collections with these properties iid's (independent identically distributed variables).

2.1.2 Mean, covariance and correlation

Given a dataset $\mathbf{x}_1^T, \mathbf{x}_2^T, \dots, \mathbf{x}_n^T$ of randomly distributed variables each with m features, there are two particularly important properties that describe the data, the mean and the variance (or covariance for the multivariate case). The mean is essentially the

'average' result of all observations. It is calculated by summing the values of each observation and dividing by the total number of observations. We typically denote the mean by $\bar{x}$:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where the sum above represents vector addition (component-wise). In the univariate case, the variance measures how spread out the observations are. In the multivariate case, we compute the variance of each m features, as well as the covariance between each of the features (how much two features vary with each other). This therefore means that the covariance is an $m * m$ matrix. The covariance matrix Σ is calculated as Ross, 2017, p. 235:

$$\Sigma = \sigma_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$$

The correlation matrix is the standardized covariance matrix defined by

$$\rho_{jk} = \frac{\sigma_{jk}^2}{\sqrt{\sigma_{jj}^2 * \sigma_{kk}^2}} = \text{Cor}(x_j, x_k)$$

Where x_j and x_k represent any two of the m features. This is a symmetric matrix as $\text{Cor}(x_j, x_k) = \text{Cor}(x_k, x_j)$, and the values range from -1 to 1:

$$-1 \leq \rho(x_j, x_k) \leq 1$$

The closer $\rho(x_j, x_k)$ is to -1, the stronger the negative relationship between the features. The closer $\rho(x_j, x_k)$ is to 1, the stronger the positive relationship. If $\rho(x_j, x_k)$ is close to 0 then it implies that the variables are independent of each other. Figure 2.1 shows some examples of random variables with different values of Pearson's correlation coefficients ($\rho(x, y)$) created in R.

2.1.3 Normal distribution

The normal distribution is an example of a distribution for a continuous random variable. It is one of the oldest and most commonly used distributions in statistics. The normal distribution is dependent on two parameters, the mean and standard

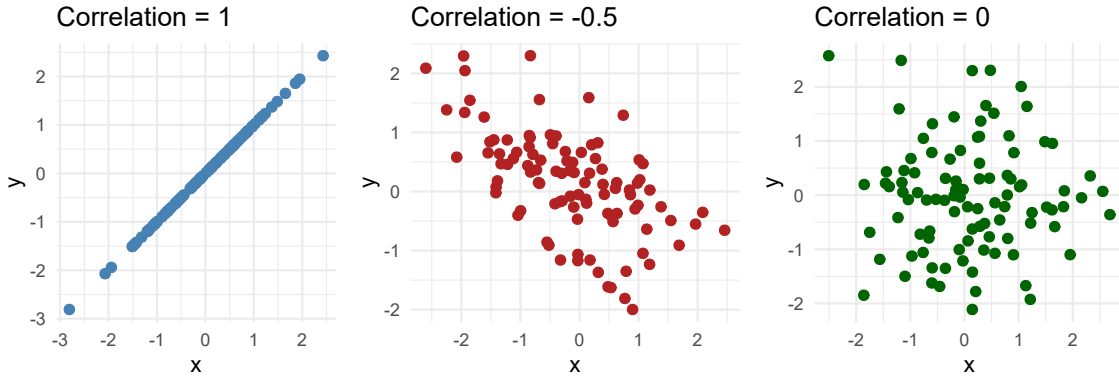


Figure 2.1: Pearson correlation coefficient examples

deviation. Given two values for these, say μ and σ respectively, the normal distribution is defined by the following probability density function (Ross, 2017, p. 267):

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

This represents a univariate variable, all values here are scalars. The pdf tells us the probability that the random variable will take value less than x . The normal distribution is recognized by its unique bell-shaped curve centered at μ . This can be seen in Figure 2.2. A Normal Distribution curve is sometimes referred to as a Gaussian curve. This distribution is a generalization of the standard normal distribution which assumes a mean of 0 and standard deviation of 1.

2.1.4 Student's t-distribution

The Student's t-distribution is another type of continuous random variable that we will come across in this project. This distribution is characterized by the following pdf (Rice and Rice, 2007, p. 193):

$$f(t; \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad (2.2)$$

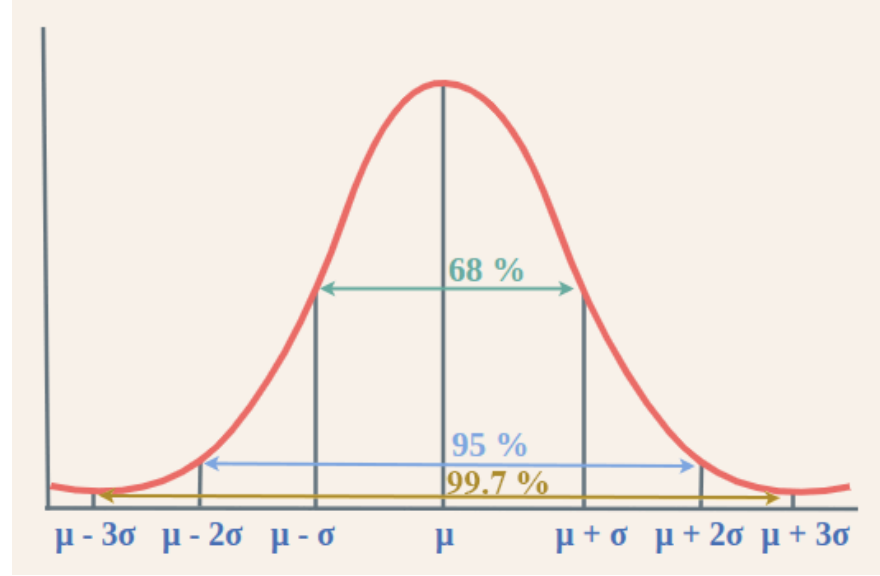


Figure 2.2: Normal distribution curve

again these are all scalars. Here ν means the degrees of freedom, a term which represents the maximum number of independent values, which may vary in a data sample. A simplified estimate for the degrees of freedom is simply the number of observations minus one, $\nu = n - 1$. In equation 2.2, Γ denotes the gamma function (Olver et al., 2010, p. 136):

$$\Gamma(z) = \int_0^{\infty} t^{z-1} \exp(-t) dt$$

This can be calculated explicitly using some interesting properties of the function, for example we have $\Gamma(n + 1) = n!$ (where $!$ represents the factorial function) (Olver et al., 2010, p. 137). The student t distribution is a generalization of the normal distribution. As $\nu \rightarrow \infty$ the student t distribution converges to a normal distribution (Ahsanullah et al., 2014, p. 53). The student t distribution is still valid when the sample size is small, and therefore ν is small, which is not the case for the normal distribution.

2.2 K-means clustering

Data clustering is an important area of research with applications in statistics, data mining, computer science, pattern recognition, image processing, marketing, etc. It is an unsupervised method where a multidimensional data set is classified into groups (clusters) such that points in the same cluster have the greatest similarity, and points

from different clusters, the greatest dissimilarity (Oti et al., 2021, p. 65). Clustering methods can be split into two different types; hierarchical and non-hierarchical (partitioning). Hierarchical methods group similar points into clusters, either by merging similar clusters together (agglomerative methods) or by splitting larger clusters (divisive) (Oti et al., 2021, p. 65). These steps are decided based on some measure of similarity between the clusters. Partitioning methods start with some initial partition of the data set and look to improve the partition step-by-step. K-means is one of the oldest and most popular examples of a partitioning clustering method.

The k-means algorithm takes a predetermined value for the optimal number of clusters, k , and searches for an optimal partition of the data, $\eta = S_1, S_2, \dots, S_k$, into k clusters. Suppose we have a data set $\mathbf{x}_1, \dots, \mathbf{x}_n$ of n data points that each have m features. Given a set value for k , the aim of the algorithm is to satisfy the following conditions (Mucherino et al., 2009, p.48):

$$\{\mathbf{x}_1, \dots, \mathbf{x}_n\} = \bigcup_{i=1}^k S_i, \quad S_i \cap S_j = \emptyset, \quad 1 \leq i \neq j \leq k.$$

This is simply a way to formally state that we are looking for a partition of $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ into k disjoint clusters. The clusters contain samples with some similarity, simply meaning close in Euclidean distance of all features. The k-means algorithm requires a measure of similarity between individual samples and clusters as a whole. To do this, we calculate the cluster mean and the distance from a point to this mean. For a given cluster S_j , its mean is calculated using:

$$\mathbf{c}_j = \frac{1}{n(S_j)} \sum_{i=1}^{n(S_j)} \mathbf{x}_{j(i)}.$$

Here, by $n(S_j)$ we mean the number of samples contained in the cluster S_j . The index $j(i)$ represents the i 'th sample in the j 'th cluster (Mucherino et al., 2009, p. 48), which is a vector with m features. The vector $\mathbf{c}_j$ is a m -dimensional and contains the means of the group for all the m features. Initially, the points are randomly assigned to one of the k clusters. Then the cluster means can be calculated. The algorithm updates the clusters in such a way that minimizes the *error function* (Mucherino et al., 2009, p. 49):

$$f(S_1, S_2, \dots, S_k) = \sum_{j=1}^k \sum_{i=1}^{n(S_j)} d(\mathbf{c}_j, \mathbf{x}_{j(i)})$$

where $d(\mathbf{c}_j, \mathbf{x}_{j(i)})$ represents the Euclidean distance between a sample $\mathbf{x}_{j(i)} \in S_j$ and the cluster mean $\mathbf{c}_j$ (Mucherino et al., 2009, p. 49). So, f records the sum of the distances of each point from the mean of the group to which it is assigned. Since the distances are always greater than or equal to zero, to minimize f we need to minimize each term $d(\mathbf{c}_j, \mathbf{x}_{j(i)})$. The algorithm achieves this by computing the distance between a sample from cluster j , $\mathbf{x}_{j(i)}$, and the cluster mean of all other clusters. If a different cluster is found, say $S_{j'}$, such that $d(\mathbf{c}_{j'}, \mathbf{x}_{j(i)}) < d(\mathbf{c}_j, \mathbf{x}_{j(i)})$ then the point $\mathbf{x}_{j(i)}$ is moved to the cluster $S_{j'}$. This will therefore reduce the error function. This process is done for all points in all clusters until an improved partitioning is created. Repeating this process iteratively eventually results in the optimal partition and the k-means clustering is complete.

We should note that to carry out this algorithm we needed to start with some value of k . It is important to choose the right value for k here, in order to gain the greatest amount of information from the clustering. The method we will use to determine the optimal number for k is called the 'elbow method'. This method requires running the algorithm for various possible values of k , then computing the sum of the squared errors within a cluster. By this we mean (Cui et al., 2020, p. 6) :

$$\sum_{j=1}^k \sum_{i=1}^{n(S_j)} \sqrt{\sum_{l=1}^m (x_{il} - c_{jl})^2}$$

For each cluster, we sum the squared distances of each sample in the cluster to that cluster mean. This is done by taking the sum of the squares of the differences between each individual scalar feature, then taking the square root to find the Euclidean distance. We do this for all samples in the cluster, then do the same for all other clusters. We then plot the total sum of these squared errors against k for each k we try. This graph will have a negative gradient, as increasing the number of clusters decreases the distance between samples and their cluster means. Whilst a lower sum of squared errors means better defined clusters, we do not want to over fit the data and use too many clusters. Therefore, we choose the value of k where the gradient of this graph starts to tail off. This point is known as the elbow of the curve, hence the name 'elbow method'.

2.3 T-distributed stochastic neighbor embedding

T-Distributed stochastic neighbor embedding is a dimensional reduction method used in data science. Dimension reduction provides a useful way to visualize patterns in complex data sets. The aim of these methods is to translate high-dimensional data sets (data that contain many features) into a lower-dimensional data set that can be plotted in two or three dimensions. The goal is to do this in a way that preserves the structure of the data. In other words, the aim is that samples that are similar to each other in the original data set should remain similar in the translated data set.

T-SNE, for short, is a nonlinear dimensional reduction method, which is different from traditional reduction methods such as principal component analysis which is linear. The aim of these linear reduction methods is to keep dissimilar points far apart in the translated data set. For high-dimensional data, the nonlinear approach of keeping similar data points close together can be a more beneficial way to maintain the structure of the data (Maaten and Hinton, 2008, p. 2580).

To see how this works, let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ denote a high-dimensional data set. We will let $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$ denote the translated form. T-SNE starts by describing the distance between two points $\mathbf{x}_i$ and $\mathbf{x}_j$ as conditional probabilities that represent the similarities between the points. The conditional probability, denoted $p_{j|i}$ is the conditional probability that $\mathbf{x}_i$ would pick $\mathbf{x}_j$ as its neighbor if neighbors were chosen in proportion to their probability density under a Gaussian centered at $\mathbf{x}_i$. Formally, this proportion can be seen as (Maaten and Hinton, 2008, p. 2581):

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)}$$

where σ_i^2 represents the variances of the Gaussian at $\mathbf{x}_i$. These variances are chosen, and there is unlikely to be one variance that is optimal for all points. Points in less dense regions will need a higher variance than those in dense regions. The $\|\cdot\|$ above represents the Euclidean norm between two vectors. We can compute a similar conditional probability to represent the distance between two points $\mathbf{y}_i$ and $\mathbf{y}_j$ in the lower dimension data set, which we denote $q_{j|i}$. For this data set, we can set the variance of the Gaussian to $\frac{1}{\sqrt{2}}$ for all i which means the conditional probability for

$\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$ becomes (Maaten and Hinton, 2008, p. 2581):

$$q_{j|i} = \frac{\exp(-\|\mathbf{y}_i - \mathbf{y}_j\|^2)}{\sum_{k \neq i} \exp(-\|\mathbf{y}_i - \mathbf{y}_k\|^2)}$$

For both of these probabilities we only care about pairwise similarities so we let $p_{i|i} = q_{i|i} = 0$, for all $1 \leq i \leq n$. If $\mathbf{y}_i$ and $\mathbf{y}_j$ have the same similarity as $\mathbf{x}_i$ and $\mathbf{x}_j$ then $p_{j|i}$ will be equal to $q_{j|i}$. T-SNE uses this fact to try and find a translation $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$ that minimizes the difference between $p_{j|i}$ and $q_{j|i}$ for all $i, j \in \{1, 2, \dots, n\}$.

T-SNE does this by minimizing the Kullback-Leibler divergence between the joint distributions P and Q (which contain the information for all $p_{j|i}$'s and $q_{j|i}$'s). We call the Kullback-Leibler divergence the cost function as we are trying to minimize it to obtain the optimal translation (Maaten and Hinton, 2008, p.2581):

$$C = KL(P, Q) = \sum_i \sum_j p_{ij} \log(p_{ij}/q_{ij})$$

t-SNE is an improved version of the standard SNE dimension reduction. The standard SNE minimizes the Kullback-Leibler divergence of the conditional probability distributions P_i and Q_i for each i individually, $i \in \{1, \dots, n\}$. T-SNE uses joint distributions, which means that the cost function is symmetrized $p_{ij} = p_{ji}$. It also uses a Student-t distribution rather than a Gaussian to compute the similarity between points in $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$. The similarities using these joint distributions are computed by the following, for p_{ij} (Maaten and Hinton, 2008, p. 2582):

$$p_{ij} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma^2)}{\sum_{k \neq l} \exp(-\|\mathbf{x}_k - \mathbf{x}_l\|^2/2\sigma^2)}$$

And for q_{ij} :

$$q_{ij} = \frac{\exp(-\|\mathbf{y}_i - \mathbf{y}_j\|^2)}{\sum_{k \neq l} \exp(-\|\mathbf{y}_k - \mathbf{y}_l\|^2)}$$

A problem with these definitions arises when we have outliers. If $\mathbf{x}_i$ is such an outlier, then $\|\mathbf{x}_i - \mathbf{x}_j\|$ is very large for all $j \neq i$. p_{ij} is then very small, which means that the position of $\mathbf{y}_i$ has little effect on the cost function. Therefore, the position of $\mathbf{y}_i$ is not well-determined by the position of the other points. We improve this by making the joint probabilities in high-dimensional space, p_{ij} , the symmetrized conditional probabilities, $p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$. Doing this means $\sum_j p_{ij} > 1/2n$ for all j . Hence, all

samples contribute to the cost function and the position of all points $\mathbf{y}_i$ will depend on their location compared to the other points, including the outliers.

Chapter 3

Data

Football data is a rapidly growing industry, but the idea has been around for a long time. As early as the 1950's Charles Reep collected football statistics by hand to suggest that "the key to scoring goals and winning games was to transfer the ball as quickly as possible from back to front" thereby indirectly starting the long-ball movement in English football (Cintia et al., 2015, p. 1). Since then, people have collected data from football matches to try and gain any type of advantage, but only in recent years has this field seen substantial growth. Now many of the top clubs have teams of data analysts dedicated to football data science, for example City Football Group and Fenway Sports Group both have data departments which provide insights for Manchester City and Liverpool respectively. Advancements in this field are largely due to the new superior data collection methods. These methods collect data of two separate types: (1) event logs, which record all events that take place in a match, (2) tracking data which uses videos or GPS to track players' position on the pitch (Gudmundsson et al., 2017, p. 5). Both of these data-types are referred to as 'Spacio-Temporal' as every piece recorded contains a time stamp and the location on the pitch where it occurred. These methods are either fully or partially automated, so they record far more information than would be possible by hand. Event logs contain over 3000 events per game and the tracking data contains around one million tracking locations per game, recording player positions 25 times per second for all 22 players (Statsbomb, 2025). In this project we delve into a large events data set from (Statsbomb, 2025). This is one of the largest football data sets ever made accessible to the public, containing millions of events obtained from

matches from 74 different competitions.

3.1 Soccer logs

As we mentioned above, the groundbreaking methods of producing large datasets for each game from automated processes have revolutionized the field of football data. The logs that identify and record all events from a game are known as 'Soccer Logs'. In this section, we look at how these logs can be useful and two different ways they can be produced.

Similar to the 1950's when Repp attempted to show football is about moving the ball forward with pace, modern day football statistics is all about trying to find data driven insights. Advancements in data science and machine learning in the 21st century allow statisticians to extract vast amounts of information from large data sets. These methods were used to great effect in industries such as finance, banking, and insurance. However, given a large enough data set, these methods can be applied to any type of data. Given football clubs intense desire to gain competitive edge, and the wealth available at their disposable, hiring teams of data scientists seems to be an inevitable path most top clubs will go down.

This therefore created a need for large data sets that contain all the information about what happens in each match. One way to do this was to find a way to record all events that happen in a match, e.g. passes, goals, shots, dribbles, and bookings. Since this would not be achievable by hand, the need to create automated computer systems that record all of these events for us arose. One approach data scientists took to create such a system was to create a program that could identify these events automatically if given a series of videos from a match. Early attempts to create such a system used Bayesian networks and the hidden Markov model (Rongved et al., 2021, p. 1034).

In 2018, a team of statisticians at the University of Liege attempted to produce a benchmark open source data set for football video analysis (Giancola et al., 2024). Giancola and his team noticed this need for an automated event detection system in football. They identified that the problem consisted of identifying unique actions in a match and assigning them an accurate time stamp. Their initial data set consisted

of 500 matches from the following leagues: Serie A, La Liga, the Bundesliga, Ligue 1, the Premier League, and the Champions League. Their original system focused only on the temporal localization of goals, cards, and substitutions. However, it is from this simple system that most of today's automated event identifiers have been developed. This was the benchmark that has inspired more than 60 new systems in the last six years alone (Giancola et al., 2024). The most advanced of these now identifies around 25 different types of events and records over 150 features for all of these types combined.

3.1.1 Automated collection methods

The initial system produced in 2018 was called SoccerNet(v1) (Giancola et al., 2024). The goal was to take a set of N untrimmed videos $V = \{v_1, v_2, \dots, v_n, \dots, v_N\}$ and extract all K actions $A^n = \{a_1^n, a_2^n, \dots, a_K^n\}$ for each video v_n (Giancola et al., 2024, p. 5). Actions contain a class type and a time stamp, where the class types were stated above (goal, card, or substitution).

The action spotting method M was defined as a function $M : v^n \rightarrow \hat{A}^n = \{\hat{a}_1^n, \hat{a}_2^n, \dots, \hat{a}_K^n\}$. This function predicts all events from the set of videos, containing a predicted class $\hat{c}$, and a timestamp $\hat{t}$. An optional confidence score $\hat{s} \in [0, 1]$ is also recorded. Generally, these action spotting methods are a composite of three primary component functions, referred to as: (1) B the backbone, (2) N the neck, (3) H the head (Giancola et al., 2024, p. 7).

The backbone gets the core components for feature extraction. Defined as $B : v^n \rightarrow F^n$ where $F^n = \{f_1^n, \dots, f_k^n\}$ is a set of k features from the video v^n . These features contain all the essential visual information from the video, forming a foundation for action spotting (Giancola et al., 2024, p. 8).

The neck then takes these features and transforms them into a more discriminative form to allow action spotting. The neck is defined as $N : F^n \rightarrow \hat{F}^n$, where $\hat{F}^n$ contains these transformed features which have also been resized, to be inputted to the head function in the next step. This process is done using principal component analysis (Giancola et al., 2024, p. 9).

The head is the final step where the event identification and temporal localization takes place. This function takes the refined features and predicts the set of actions,

formally defined as $H : \hat{F}^n \rightarrow \hat{A}^n$. Taking the composite of the 3 functions gives us:

$$M = H \circ N \circ B : v^n \rightarrow \hat{A}^n$$

our event identifier function M which predicts a set of events from each given video.

3.1.2 Tagging methods

The other method used to create event data involves professional data collectors known as operators who use proprietary software to log events known as a tagger. The operators use the tagging software to manually tag each event they observe, and the tagger then creates a new event on the timeline each time. In general, three operators work on each match to tag all events that occur (one operator per team and one as a supervisor for the whole match). Firstly, the operators initialize the team formations, and positions and shirt numbers for each player. Then for each touch of the ball, the operator selects the player and creates a new event. They select the event type (shot, pass, tackle) and subtypes which are different for each event type (e.g. goal or wide for shot outcome). Finally, they add the coordinates of the event on the pitch (Pappalardo et al., 2019, p. 2).

After the tagging is completed, a quality control procedure is performed. Firstly, an algorithm is run to remove any obvious errors from the operators. This algorithm includes steps such as checking that the events recorded by both operators match up, identifying any impossible sequence of events, and spotting any events that have been missed. There is also a human quality control step for randomly selected matches. This involves inspecting all events to ensure they have correct type and subtype (Pappalardo et al., 2019, p. 2). This tagging method is how Statsbomb collected the data we will use in this project.

3.2 Uses

We have outlined two ways of collecting large amounts of event data from football matches. Given these data sets, we can use them to provide value in many different areas of football. These can include insights that help clubs, players, media teams,

and even referees. Here are some brief overviews of how we can extract value from these data sets:

3.2.1 Player comparison

We can use the event logs to compare players. Depending on the types of player we want to compare, we can filter the data to contain only the events that are important to these types of player (e.g. tackles, interceptions, clearances for defenders). We can then use various methods to see which players are the highest performers in each of these metrics. This can be helpful for clubs when it comes to recruiting new players.

3.2.2 Team tactics

The tracking data can help coaches understand team tactics regarding formation shapes. We can use clustering methods to see players average positions in a game relative to each other. This lets us visualize the formation that teams use. Clubs can use this to identify the tactics that their opposition utilize. They could theoretically then see which team shapes had the best success against this opposition by using the data from all previous matches against this team. Knowing this, managers can set their teams up in a formation that gives them the best chance of success.

3.2.3 Game modeling

We can use the data to create models to better understand the game. For example, expected goals models which consider all aspects of a shot to predict the probability of a goal being scored. We can model player value by measuring the value each of their actions in a game give to their team, and summing to see their net value added to the team.

3.2.4 Explaining ideas to players

Coaches can use visuals and insights from the data to explain their ideas to players in a simpler way. For example, a manager may want the players to pass the ball into the box more often. We may be able to use statistical models to show that the more a

team passes into the box, the higher their expected goals. This can be a good way to explain to players the key things they need to do in a game to increase the team's chances of winning.

3.2.5 Fairer refereeing

Automated event identifiers can also help refereeing. VAR teams can use event logs while making decisions about red cards and penalties. The automated system will identify these events as penalties or cards if they are similar to examples from previous games, so referees can use this to make more consistent decisions.

As stated earlier, in this project we focus on player comparison.

3.3 Statsbomb data

3.3.1 Accessing the data

Statsbomb provides free event data for public use, collected from various different leagues over many seasons. This data set is constantly being updated with new matches that they have collected data from. The data they collect come in three different forms (Statsbomb, 2025):

1. The competitions data contains all relevant information about the competitions the data was collected.
2. The matches data contains all relevant information about the matches the data was collected from.
3. The event data contains the information describing all the events that were recorded in the matches.

To access the data, Statsbomb created a package in R which we use to load the data. We install this package using dev tools and github, the package is called "statsbomb/StatsBombR". Once this is installed, we have access to the following functions that load the data (Statsbomb, 2025):

- `FreeCompetitions()`: This loads all the competitions from which we have matches data.

- `FreeMatches()`: This loads all matches from a given set of competitions we choose.
- `free_allevents()`: This loads all the events from a given set of matches we choose.
- `allclean()`: This extracts all relevant information from the set of events we give it.

3.3.2 Competitions

The competitions data gives us information about all the competitions that currently have accessible matches. At the time of writing, there are 74 such competitions including various seasons from men's and women's competitions. In this project, we focus on the 2015/2016 Premier League season. Table 3.1 shows some summary information we have for this season and the 2015/2016 Bundesliga as another example:

season_id	competition_name	competition_gender	season_name	match_updated
27	Premier League	male	2015/2016	2024-06-12
27	Bundesliga	male	2015/2016	2024-05-19

Table 3.1: Competition information

Here we have the basic information that defines the seasons, along with the date that the matches were last updated.

3.3.3 Matches

The matches data give overview information about each match we have access to events from. Tables 3.2 and 3.3 contain the most important information we have, for five example matches from the 2015/2016 Premier League season:

The examples in these tables show information from five of Leicester City's important wins in their title-winning campaign. Table 3.2 gives an overview of some key information from the match, including the date, match week, stadium, and referee. Table 3.3 shows the teams involved and the final score. We have access to all matches played that season. This corresponds to 380 matches in the Premier League ($38 * 20/2 = 380$).

Index	Match date	Stadium	Referee name	Match week
68	2015-09-13	King Power Stadium	Mike Dean	5
101	2015-12-19	Goodison Park	Jonathan Moss	17
102	2015-11-21	St James' Park	Mike Jones	13
161	2015-12-14	King Power Stadium	Mark Clattenburg	16
190	2016-01-13	White Heart Lane	Lee Mason	21

Table 3.2: Match overview examples

Index	Home team name	Home score	Away team name	Away score
68	Leicester City	3	Aston Villa	2
101	Everton	2	Leicester City	3
102	Newcastle	0	Leicester City	3
161	Leicester City	2	Chelsea	1
190	Tottenham Hotspur	0	Leicester City	1

Table 3.3: Match score examples

3.3.4 Events

The event data contain all the events recorded from a collection of matches. For each match, over 3400 events are logged. This means that for that Premier League season, we have access to more than 1,290,000 events. This event data set is therefore vast, containing more than 180 features for the various events. The analysis in this project will be based on the data in these event data sets. Table 3.4 describes the key data recorded for each event:

3.3.5 Event types

For each event, additional information is recorded relating to that unique event type. Table 3.5 shows some of the key event types we will focus on in the project, and the additional information recorded.

For a full list of the event types, see Statsbomb's github page in the appendix. This also contains more additional information recorded for passes and shots, and figures showing the pitch coordinates used and position abbreviation guides.

Feature	Description
id	A unique identifier for each event.
index	A chronological order index for each event in a specific match.
period	Specifies the half, 1 for first half, 2 for second.
timestamp	Exact time on match clock the event took place, recorded to the millisecond.
possession	Index describing which possession of play the event is from. Every time a new team gains control of the ball the index increases by 1.
possession team	The team in control of the ball.
play pattern	"Regular play", "Corner", "Free kick", etc.
team id/name	id/name of team event is associated with
player id/name	id/name of player event is associated with
position	Abbreviation of players position. See list of abbreviations in appendix.
location	[x,y] coordinates localizing the event on the pitch. See appendix for coordinates of key points on pitch.
under pressure	Boolean valued feature, TRUE if team is under pressure.
out	Boolean valued feature, TRUE if event results in the ball leaving the pitch.
related events	List of event id's that are related to this event. E.g. a shot might be related to a block or a save.
type	The type of event e.g. shot, block, tackle.

Table 3.4: Event features description

3.3.6 Preprocessing the data

When we load the event data into R using the functions specified in 3.3.1, we create a massive table containing all the events from a set of matches. Each row in this table corresponds to an individual event such as a shot, pass, or tackle. The information in tables 3.4 and 3.5 tell us about the different features recorded for each event. Specifically, table 3.4 tells us about the features available for all event types, and table 3.5 tells us about the specialized features that are only applicable to specific types. There are 23 different types of events. Each event contains somewhere between 1–18 specialized features. This means that the event data set has a total of 186 columns. We want to perform analyses on the season as a whole, so we load the

Event type	Additional features	Feature description
50/50	outcome	"Won" or "Lost" depending on outcome.
Bad behavior	card	Color of card ("yellow", "red")
Block	deflection	Boolean, block was a deflection.
	offensive	Boolean, block led to an attack
	save block	Boolean, block saved a shot
carry	end location	[x,y] coordinates where the carry ended.
clearance	aerial won	Boolean, event was aerial.
	body part	Body part used to clear the ball ("Head")
dribble	nutmeg	Boolean, the ball went through a defenders legs.
	outcome	"complete" or "incomplete".
duel	type	"Aerial lost", "tackle", etc.
	outcome	"won", "lost out", "lost in play", etc.
interception	outcome	"won", "lost out", "lost in play", etc.
pass	recipient	Player who the pass was intended for.
	length	Length of the pass.
	end location	[x,y] coordinates pinpointing where the pass ended on the pitch.
	assisted shot id	Event id of the shot that the pass assisted, if applicable.
	cross	Boolean, the pass was a cross.
shot	goal assist	Boolean, the pass assisted a goal.
	key pass id	Event id of the pass that led to the shot.
	end location	[x,y,z] or [x,y] coordinates of shot's end location in the goal.
		See appendix for key coordinates of the goal.
	open goal	Boolean, keeper was not in a position to make a save.
	Statsbomb xg	Probability of shot resulting in a goal. xG calculated from Statsbomb's xg model.
	body part	Body part used for shot ("Head").
	outcome	Outcome of the shot ("Goal", "Saved").

Table 3.5: Key statsbomb events

events from all 380 matches, resulting in 1,313,783 total events/rows.

This table is therefore very vast and contains a copious amount of information. However, it is not easy to perform analyses with the data in this form. We wish to compare player performance based on various offensive/defensive metrics. We would therefore prefer to have tables containing player's individual statistics for the

season, such as their number of shots, goals, passes, or tackles. With the data in this form, it is easier to use our statistical methods to compare the players. We therefore wish to transform the event data, a list of individual events, into various summary datasets containing summary statistics for individual players' seasons.

The following code shows how we do this for a basic example, where we create a player summary table containing five finishing metrics (shots, shots on target, goals, shooting accuracy, and shooting success). Here, shooting accuracy means the percentage of shots that hit the target, and shooting success is the percentage of shots that result in goals. In this example, StatsBombData represents primary event data set containing all events from the 2015/2016 Premier League season.

```
#filter so its just attackers
#get shots, shots on target and goals for attackers.
att_actions<-events%>%filter(position.id==17 |
position.id==21 | position.id==22 | position.id==23 |
position.id==24 | position.id==25)%>%group_by(player.name,
player.id)%>%summarise(shots=sum(type.name=="Shot",
na.rm=TRUE),
SoT=sum(shot.outcome.name=="Goal", shot.outcome.name=="Saved"
, shot.outcome.name=="Saved To Post", na.rm=TRUE),
goals=sum(shot.outcome.name=="Goal", na.rm=TRUE))

#add minutes played to table
att_actions=left_join(att_actions, player_minutes)

#get 90s
att_actions = att_actions %>% mutate(nineties = minutes/90)

#transform features to be per 90 mins, and get shot accuracy
#and shot success.
att_actions=att_actions%>%mutate(shot_accuracy=SoT*100/shots
s, shot_success=goals*100/shots,
shots_per90=shots/nineties, SoT_per90=SoT/nineties,
```

```
goals_per90=goals/nineties)
```

```
#omit strikers with 0 shots
```

```
att_actions<-na.omit(att_actions)
```

```
#filter so only care about players with more than 5 matches played
```

```
att_actions=att_actions%>%filter(nineties>5)
```

The `'%>%'` function above is from the Statsbomb R package and allows us to filter/mutate the data. Firstly, we used the "filter" function in R which removes any events from players who do not play in relevant positions. For this example, we keep only events from strikers and attacking midfielders, who's finishing ability we are interested in. We will use this function in a similar way in all other examples. We then use the "group_by" function to group events based on their `player_id`. We do this in each example as we can then use the "summarise" function which creates a new table containing features about each player individually. In the "summarise" function, we first create a column called shots. We use the "sum" function which counts the number of events whose `event_name` is shot for each player and returns the tally. We use the same function when creating the column for shots on target, however, here we count events who's `event_name` is shot, and their `shot_outcome_name` is some outcome that corresponds to a shot on target. For goals, we count all events where the `event_name` is shot, and `shot_outcome_name` is goal. For other examples, we will use this sum function similarly to count all events that correspond to a feature we are interested in.

We want these features to be per ninety minutes rather than the raw tally, so a players playing time does not affect our perception of their performance. We use the "get.minutesplayed" function from Statsbomb's R package, which creates a table of player minutes. We then use the "left_join" function which adds the player's minutes to our summary table. The "mutate" function allows us to add an extra column to the table for their nineties, which is simply the players' minutes divided by ninety (as there are ninety minutes in a game). We use "mutate" again to get our features in terms of per ninety minutes, by dividing the original features by the `nineties` column. In this line of code, we also create the `shot_accuracy` and `shot_success`

columns by dividing the number of shots by the number of shots on target, and number of goals respectively. We finally use the 'filter' function to filter the data to only contain players that have played a minimum of five games. This eliminates anomalies that did not play enough. Before performing the clustering, we also use the "subset" function to remove all columns that we do not need in the clustering such as `player_id` and raw numbers of shots and goals. All our other examples use similar functions in this way to transform the raw events data set into a player summary statistic table.

Chapter 4

Results and Discussion

4.1 Player comparison clustering and t-SNE

In this section, we attempt to use k-means clustering to identify groups of players that perform at higher levels than others and determine what exactly makes them different from the rest. By doing this, we can find qualities in players that are valuable, which might not be obvious to see when watching the game. We use t-SNE here as a visual aid.

4.1.1 Finishing ability

Firstly, we focus on the finishing ability of the players (efficiency of their shots). To do so, we start by filtering the data set so that it contains only events from attackers. By attackers, we mean positions 17 and 21–25 inclusive in the Statsbomb position guide found in the appendix. We then produce a summary table that records the number of shots per game, shots on target per game, goals per game, shot accuracy, and shot success for each of the attackers. Shot accuracy means the percentage of shots that hit the target (finish somewhere within the frame of the goal). Shot success is the percentage of shots that the outcome is a goal. Now that we have this table, we can use our clustering methods to identify groups of players that have similar values for each of these metrics.

Before we do the clustering, we should plot the correlations of each of our features to understand the relationship between them.

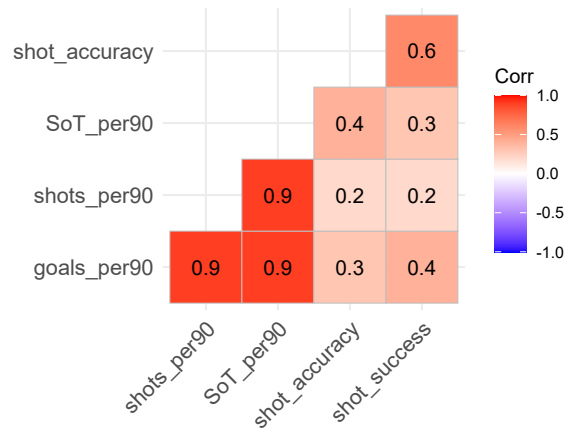


Figure 4.1: Correlations of finishing features

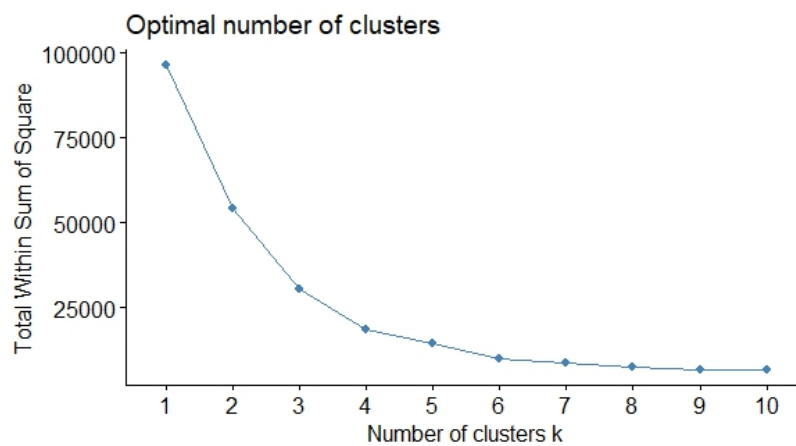


Figure 4.2: Sum of squared differences for goal scorers

Figure 4.1 plots these correlations. Instantly we notice that the goals per 90, shots per 90 and shots on target per 90 are all highly correlated(0.9). This makes sense as the more shots you take, the more chances you have to score or hit the target, so we expect there to be a high correlation between these. Shot success is also positively correlated to shot accuracy, which makes sense as you cannot score unless you hit the target. All other features have a low positive correlation. The correlation between shots per 90 and both shot accuracy and shot success is only 0.2 so these behave more or less independently of each other.

We can now perform the clustering. Using k-means clustering, we first need to identify the optimal number of clusters. We do this using a sum of squared distances plot shown in Figure 4.2:

Analyzing the plot above tells us that the optimal number of clusters for these

data is four, as this is where the curve's "elbow" is located. We now attempt to cluster the players using k-means with four centers. Table 4.1 shows the cluster means for each of the features we stated (*per 90 minutes):

Cluster	Shot Accuracy(%)	Shot Success(%)	Shots*	Shots On Target*	Goals*
1	33.397	10.732	1.581	0.539	0.186
2	54.053	13.295	1.563	0.769	0.251
3	4.293	0.687	0.273	0.026	0.003
4	100.000	83.333	0.085	0.085	0.073

Table 4.1: Finishing cluster means

The cluster sizes are 91, 37, 34 and 3. Notice that the fourth cluster only has three players. All of these players had 100% shot accuracy, but this is because they took very few shots. We should therefore ignore these players and class them as outliers.

Considering the other three clusters, we observe that the first two contain players that are relatively proficient in front of goal, whilst the third cluster contains players who are inefficient in front of goal. Analyzing the group means for cluster 3, we see that the players in this group take fewer shots and have poor shooting accuracy, so it is no surprise that their mean goals per game are low (0.0027).

Comparing the first and second clusters, we see that players in both clusters take on average a similar number of shots, around 1.57 per 90 minutes. These two groups contain the best goal scorers, scoring 0.18 and 0.25 times per match, respectively. Interestingly, the players in cluster 2 score significantly more goals per game than those in cluster 1 despite both taking a similar number of shots. It appears that this difference is due to the superior shooting accuracy of players in cluster 2. Players in group 2 average 54% shooting accuracy compared to 33% for players in group 1. This result confirms a common coaching principle that states the importance of hitting the target. It suggests that players who hit the target will score more frequently, so this is a key trait that should be considered when scouting goal scorers. This is a trivial insight, however, it shows us that clustering players based on their summary data works and gives us insights about the game that are backed up by general football understanding.

Looking at the clusters to which each player was assigned, we can verify that our clustering makes sense and is valid. According to the group means, the players in

cluster 2 are the best goal scorers. We would therefore hope that the most successful strikers were assigned to cluster 2. Harry Kane, Marcus Rashford, Jamie Vardy, and Sergio Aguero (four of the leagues best strikers) as expected are in cluster 2. Cluster 1 also has some good goal scorers, Wayne Rooney, Alexis Sanchez, Eden Hazard, Graziano Pelle, and Emmanuel Adebayor, to name a few. This makes sense as we identified cluster 1 to be good goal scorers too, just slightly less efficient.

Using the Statsbomb package in R, it is possible to make a graph that creates a visual representation of a football pitch. We can then plot specific events on the pitch using their localization coordinates (Statsbomb, 2025). We can do this with players' shots and plot each shot a player took over the season on the pitch. We do this now with one player from each cluster, so we can visualize their shots from the season to better understand the differences. For this purpose, we choose Emmanuel Adebayor as a representative for cluster 1, Jamie Vardy for cluster 2, and Moussa Sissoko for cluster 3.

The shot maps for each of these players are shown in the figures 4.3 and 4.5, and 4.7. The color of each point represents the Statsbomb xG value for the shots. This value measures the probability of a goal being scored from this position. It takes into account the distance from the goal, the body part used, the pressure the player is under, and the angle of the shot. The shape of the points shows the body part used and if a goal was scored, the points are highlighted red. Adebayor had no shots with his left foot that season.

Note that Adebayor played much fewer games than Vardy and Sissoko, this is why he took fewer shots. He took 1.62 shots per game, which is a normal amount for someone in group 1. Jamie Vardy was one of the best strikers that season, so his shot map is above average for someone in cluster 2. He took 3.1 shots per game, which is considerably higher than the group 2 mean. Sissoko took 0.52 shots per game, with 20% shot accuracy, so he was one of the higher performers for cluster 3.

Vardy took more shots per game than Adebayor, but he also scored a higher percentage of his shots. Vardy's shot accuracy was 43% while Adebayor's was 33%. This was a key difference we saw between the means of clusters 1 and 2. Analyzing the shot maps lets us visualize the effect this has. We see that eight of Adebayor's shots were very close to the goal and had relatively high xG values. However, only

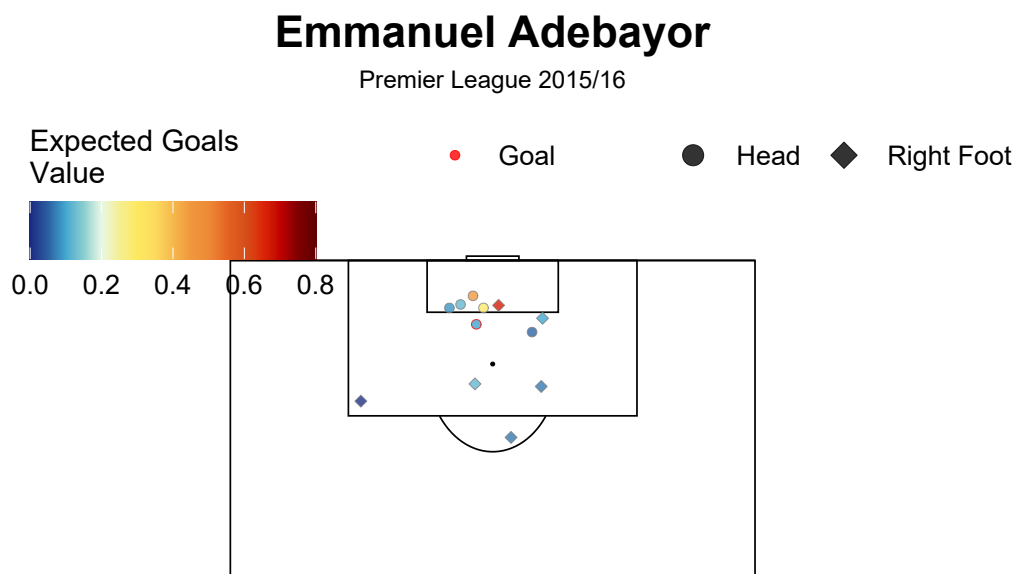


Figure 4.3: Emmanuel Adebayor shot map



Figure 4.4: Emmanuel Adebayor, 2015/2016

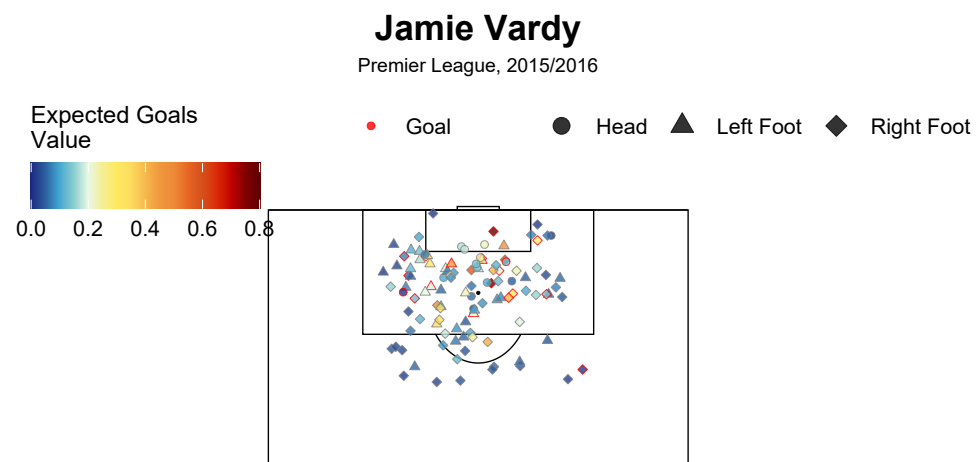


Figure 4.5: Jamie Vardy shot Map



Figure 4.6: Jamie Vardy, 2015/2016



Figure 4.7: Moussa Sissoko shot map



Figure 4.8: Moussa Sissoko, 2015/2016

one of these shots resulted in a goal. Around a third of Vardy's shots from similar positions resulted in a goal, likely because Vardy was getting more of these shots on target.

Sissoko's shot map is what we would expect from a player in cluster 3. He played a similar number of games to Vardy, 37 compared to 36 for Vardy, but took far fewer shots. His shots were less accurate, which meant that he only scored one goal from his 38 shots.

Finally, we wish to visualize the clustering. To do this, we produce a t-SNE plot in R. This plot takes the information from the features and reduces the information into two summary dimensions, which we can plot against each other to see how the different players vary. Figure 4.9 shows this.

We can see from the plot that the players group into around ten clusters when the information is reduced to the two dimensions. If we look at the players in each small cluster in this plot, we find that they all come from the same cluster in our k-means clustering. For example, Daniel Sturridge, Danny Wellbeck, Jermaine Defoe, and Odion Ighalo are all in the same small group in figure 4.9, and they were all assigned to cluster 2 in our clustering. Maroane Fellaini, Chris Brunt, Matt Jarvis, and Craig Gardener are in another cluster in figure 4.9, and these are all players from cluster 3 in the k-means clustering. Figure 4.9 gives us a nice way to visualize the different groups of players. It agrees with the findings obtained from the clustering, so it is a good accompanying plot to have.

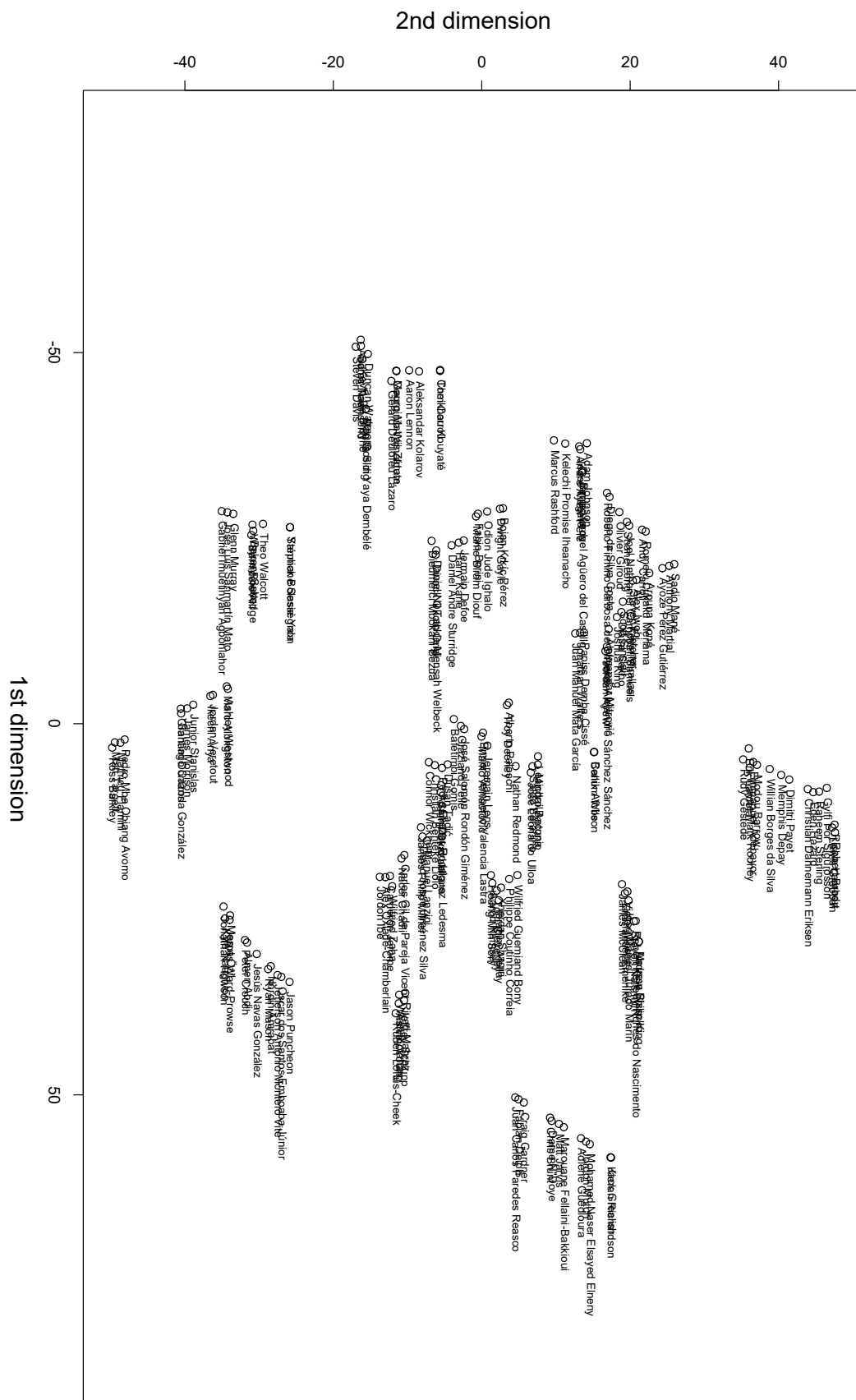


Figure 4.9: Finishing t-SNE reduction plot

4.1.2 Passing ability

In this subsection, we focus on the passing ability. In particular, we wish to compare the passing performance of all midfielders during the season. Passing is a vital attribute for midfielders as it helps teams control the game and keep possession. This is the foundation needed to create chances and score goals. We will use k-means clustering again here to identify similar groups of players based on their passing.

First, we need to prepare the data for clustering. Similarly to what we did with the finishing clustering, we start by filtering the event data so it only contains events from midfielders this time. By midfielders, we mean positions 9–16 and 18–20 in the Statsbomb position guide. We then produce a summary table containing information about the players' passes. We search for all events that have the type name "Pass" and group them by player name to get a tally of each player's attempted passes. We then search for passes with the outcome feature "Incomplete", "Out", or "Pass Offside". Counting these gives us the number of unsuccessful passes. From these we can work out the players' passing accuracy (percentage of attempted passes that were successful) and their number of successful passes per 90 minutes. We then split the passes into short, medium, and long by looking at their "pass length" feature. Here we class passes under 10 yards as short, 10–35 yards as medium, and over 35 yards as long passes. We then create a tally for short, medium and long successful passes per 90, and the accuracy of these three types of passes. Finally, we record the percentage of passes that were short, medium, and long for each player.

We now have a table containing summary statistics for the players' passes throughout the season. Before we do the clustering, we should calculate the correlations between each of these features to better understand their relationships.

Figure 4.10 shows the correlations for all the passing features that we will look at when clustering the players. We have more features here than we had for the finishing clustering, and more variation in the correlations. Inspecting the correlations we notice that there is a perfect positive correlation between pass success and medium pass success, as well as complete passes per 90 and medium passes per 90. This makes sense as the majority of passes are medium passes, so this dominates the overall pass success percentage and total pass count. We also see that there are some negatively correlated features here, unlike the finishing features, which were all

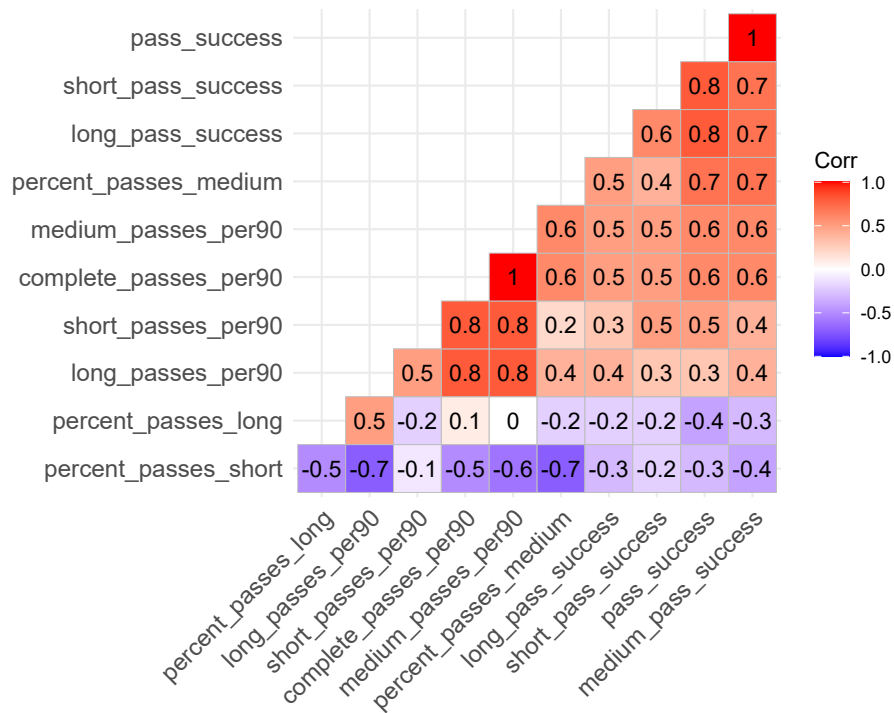


Figure 4.10: Passing features correlations

somewhat positively correlated. The percentage of passes short has some degree of negative correlation with all other features. It is understandable that this has a negative correlation with most other features, as players who tend to pass the ball shorter more often may not be as confident passers. This would suggest that their pass success rates would generally be lower and the number of passes they make lower too. The percentage of short passes is highly negatively correlated with the number of passes, medium passes, and long passes. The percentage of passes long also has slight negative correlation with the different pass success percentages, but is more or less independent to the number of passes per 90. Unsurprisingly, the number of passes, short, medium, and long, and the pass success percentages (short, medium, and long) all have varying degrees of moderate correlation with each other. Players who pass the ball more often will likely be better passers, and therefore have higher pass success percentages, so these correlations are understandable.

We now use k-means clustering to identify similar groups of players according to these passing features. Figure 4.11 shows the sum of squared differences plot for this clustering. Using this graph, we conclude that three is the optimal number of clusters as this is where the elbow is located. Table 4.2 shows the cluster means

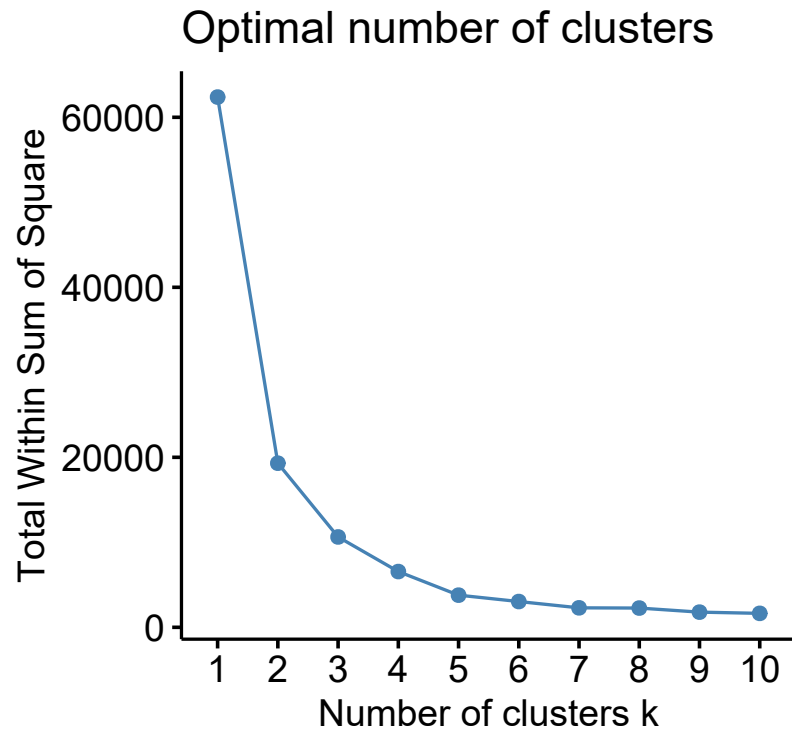


Figure 4.11: Sum of squared differences for passing clustering

obtained from the k-means clustering.

Feature	Cluster 1	Cluster 2	Cluster 3
Pass Success (%)	74.03	85.47	79.46
Short Pass Success (%)	72.95	82.26	77.26
Medium Pass Success (%)	78.77	89.28	84.02
Long Pass Success (%)	49.72	66.55	58.65
Completed Passes Per 90	13.54	49.96	32.16
Short Passes Per 90	3.69	9.06	6.64
Medium Passes Per 90	8.79	36.70	22.66
Long Passes Per 90	1.04	4.17	2.84
Percent of Passes Short	27.65	18.80	21.49
Percent of Passes Medium	61.47	70.32	66.38
Percent of Passes Long	10.81	10.84	12.09

Table 4.2: Passing cluster means

The cluster sizes were 57, 41, and 62 respectively. To understand this clustering, a key feature to analyze first is the number of passes that were completed per 90. The players in cluster 1 recorded an average of 13.54 passes per 90. Players in cluster 2 averaged 49.96 passes per 90, whilst those in cluster 3 averaged 32.16. So clearly cluster 1 contains players who do not make as many passes in the game and are

slightly less involved. Let us look at some of the players in this cluster to better understand this. Some key players in cluster 1 are the following: Anthony Martial, Heung Min Son, Jack Grealish, Alex Oxlade Chamberlain, Eden Hazard, Sadio Mane, and Wayne Rooney. We notice that these players are all very attack minded. Their job in their respective teams is to try and create chances, so it makes sense they have a lower number of passes per game because they are more direct on the ball rather than just keeping possession like deeper midfielders. Another feature that sets players in this cluster apart from the other two clusters is the percentage of passes that were short. On average, 27.65% of the passes made by the players in group 1 were short passes, compared to 18.8% and 21.49% for the players in groups 2 and 3. This elevated percentage makes sense, as attacking players have to play more short intricate passes to break down defenses. This is an interesting and useful insight, as it suggests that we can use the percentage of passes that were short as a rough guide for how much attacking responsibility any given midfielder has. This gives us a better picture of the types of passes they are making. Obviously, not all attacking midfielders will have this feature, but in general it is a good guide to use.

In contrast to cluster 1, clusters 2 and 3 contain players who pass the ball a lot more according to the cluster means for completed passes per 90. We would therefore expect to see more deeper midfielders in this role who control how their teams play with their passes. Indeed, some examples of players in cluster 2 are: Aaron Ramsey, Ander Herrera, Bastian Schweinsteiger, David Silva, Erik Dier, Michael Carrick, and Nemanja Matic. These are all the type of midfielders we would expect, pure center midfielders who are not responsible for attacking, but more focused on controlling the game with their passing. Similarly, cluster 3 contains the players: Charlie Adam, James Milner, Jordan Veretout, Matt Ritchie, Ruben Loftus Cheek, and Victor Wanyama. These players also play right in the center of the pitch, where they have to make many passes. We notice that the clusters 2 and 3 both contain this similar caliber of players, however, there are some notable differences between their cluster means. Both of these clusters contain players who play in the center of their teams, and their job is to make many passes and control how their teams play. However, players in cluster 2 make significantly more passes than those in cluster 3, they also have a notably higher pass success percentage (85.47% compared to 79.46%). Since these

contain similar types of player, it appears that being in cluster 2 or cluster 3 is a good way to differentiate between the best center midfielders (those in cluster 2) and the less impressive center midfielders (those found in cluster 3).

There are some potential exceptions to this overview of the clusters, however, for example, Mesut Ozil and David Silva were both assigned to cluster 2. In our general understanding of the clustering, cluster 2 should contain deep midfielders who do not have much attacking responsibility. However, if you have watched these players play, you know that they are heavily involved in their teams attacks. Analyzing the indicator feature we referenced earlier, we see that 28.78% of Ozil's passes were short, and Silva had a similar 28.56%. This provides evidence that these players had attacking responsibility, as these percentages are very close to the mean of group 1 (27.65%). Their average number of passes per 90 were 56.73, and 50.33 respectively, so this shows why they were assigned to cluster 2. Both of these players made a similar number of passes to the deep center midfielders found in clusters 2 and 3. However, since they have a lot more attacking responsibility compared to most players in cluster 2, this highlights how exceptional their seasons were. Their ability to control the game making that many passes, on top of the attacking responsibility they took on shows how influential they were to their teams that season. Our next section looks at the players' creative ability. I predict that these two players will be assigned to the highest performing group for creative ability. If this is the case, coupled with the fact that they appeared in the top cluster for passing ability, it will confirm to us how impressive their seasons were.

The t-SNE plot in figure 4.12 represents the players passing features compressed into two reduced dimensions. This provides a good visual understanding of the players passing statistics and this clustering. We notice there is a group of around 40 players in the bottom left section of this plot, containing players like Jack Grealish, Heung Min Son, Wayne Rooney, Memphis Depay, and Nathan Redmond. Unsurprisingly, all of these players come from the same cluster, cluster 1. These are all heavily attacking players, so it makes sense that they are grouped together in this dimension reduction plot, and they are distant from the deeper midfielders. The right central area of the plot contains three small clusters of players. Within these clusters are most of the players from cluster 2, for example: Bastian Schweinsteiger,

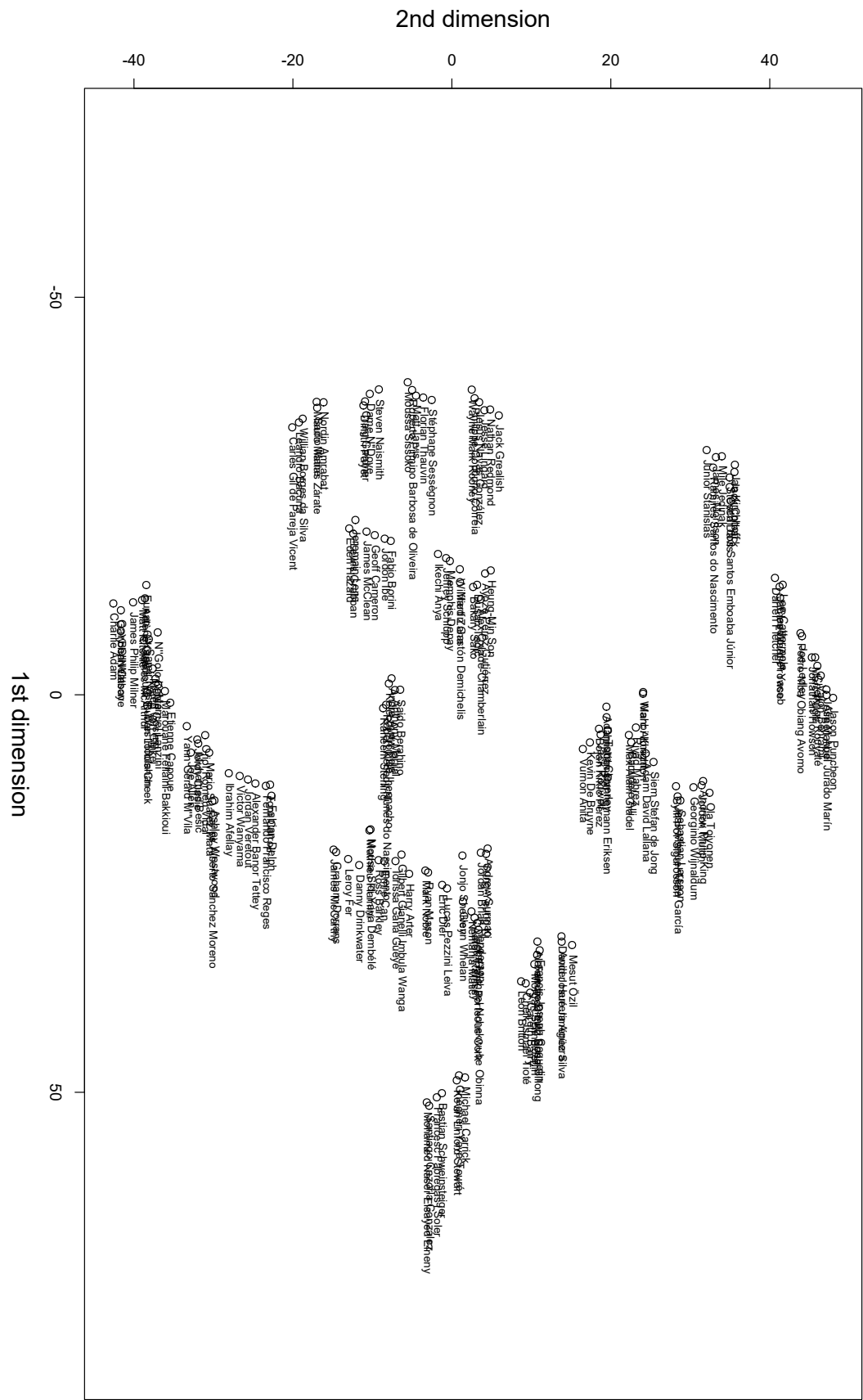


Figure 4.12: Passing t-SNE reduction plot

Michael Carrick, Harry Arter, Danny Drinkwater, and Erik Dyer. An interesting feature of this t-SNE plot is the location of the players in cluster 3. The players in this cluster appear at two opposite ends of the plot. The bottom section of the plot contains a large cluster of players, all or most of which come from cluster 3. James Milner, Victor Wanyama, Jordan Vertout, and Charlie Adam are all in cluster 3 and also in this area of the plot. Meanwhile, the top section of the plot contains a couple of clusters where various players from cluster 3, such as Georginio Wijnaldum, Junior Stanislas, Jason Puncheon, and Darren Fletcher, appear.

Looking at the player profiles in these two sections of the graph helps us to understand why players from the same cluster appear at different end of the plot. The players found at the top, like Puncheon, Stanislas, and Wijnaldum, are known to be quite attacking players. This is reflected in their percentage of short passes, which are 25.30%, 24.13%, and 30.43%, respectively. These are all higher than average for cluster 3 players and more similar to the average for cluster 1. It is no surprise therefore that some players from cluster 1, such as Christian Eriksen and Kevin De Bruyne, appear in this area of the graph. On the other hand, the players I mentioned at the bottom of the plot (Milner, Wanyama, Adam) are known to be more defensive midfielders. Their short pass percentages are 17.92%, 16.70%, and 14.51%, respectively. These are much lower than the players' at the top of the plot, so this appears to be a key feature responsible for the t-SNE reduced dimensions. These examples further back up our finding that short pass percentages are a great guide to determine how attacking a player is.

This t-SNE plot has given us a greater understanding of the clustering structure. Cluster 3 contains a mix of players similar in nature to those in cluster 2 (found at the bottom of the t-SNE plot) and others similar to players in cluster 1 (found at the top of the plot). One final note is that Mesut Ozil and David Silva are both located close to the top section of the plot, which contains creative players. This further confirms that they are similar to both players in cluster 1 and cluster 2, as they had big impacts in the passing play and the attacks too.

4.1.3 Creative ability

In the previous subsection, we identified a group of midfielders who make fewer passes because their responsibility in the team is to create chances and be more direct. In this subsection, we look at these players and other attackers' creative attributes.

By creativity, we mean a players ability to create chances like shots or goals, or general dangerous attacking situations. To measure this, we will focus on players' number of dribbles, shot assists, goal assists, through balls, crosses, passes into the box, and carries. We also look at players' numbers of miscontrols to see how often they lose the ball.

The creative responsibilities of a team are generally given to wide midfielders, wingers, and attacking midfielders, so we will only consider these players here. We start by filtering the data so that it only contains players in these positions (12, 16–21 in the Statsbomb guide). Then we group by player name and id so we can create our summary table. We search for events with `dribble.outcome.name` equal to "Complete" or "Incomplete" to get the number of successful and unsuccessful dribbles. We look for passes with boolean value `TRUE` for the `shot.assist` and `goal.assist` features to get the number of these. Similarly, to get the number of crosses we count the number of events with the `pass.cross` feature equal to `TRUE`. For through balls, we count events whose `pass.technique.name` is "Through Ball". To count the number of passes into the box, we count the number of events with type name "Pass" and end location feature between 62 and 18 on the y-axis of the pitch, and greater than 102 on the x-axis, as this area of the pitch represents the offensive penalty box. Finally, for carries and miscontrols, we simply search for events with the corresponding type names.

Figure 4.13 shows the correlation coefficients for each of the features that we use in the clustering. Here, most of the features have similar correlations with each other. Except for the dribble success percentage, all features share a positive correlation in the range of 0.5-0.9. As for dribble success, this has a low negative correlation with all of the other features, and most negative with miscontrols per 90, with a correlation coefficient of -0.4. A negative correlation between these two is understandable as better dribblers will have better control, so fewer miscontrols. The positive correlations between all other features makes sense also, as they are all

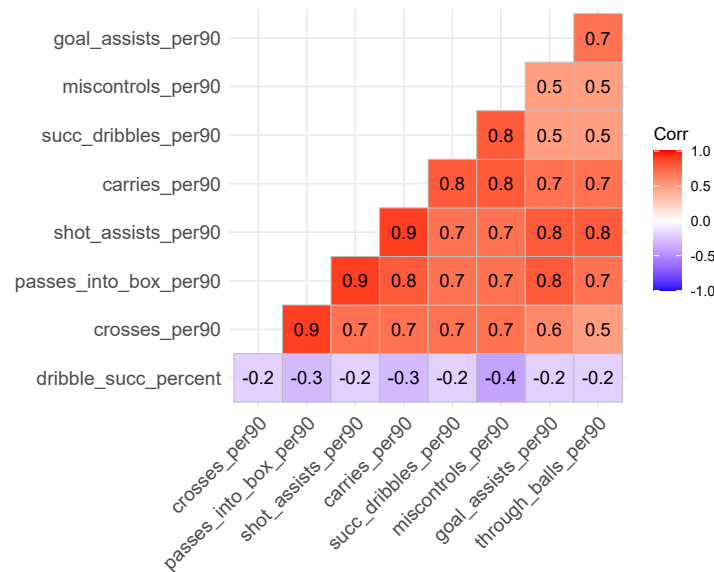


Figure 4.13: Creativity features correlations

creative metrics, we expect positive correlation between them.

Using the graph in figure 4.14, we see that the optimal number of clusters for these features is three. We should perform k-means clustering with three centers. Table 4.3 shows the cluster means.

Feature	Cluster 1	Cluster 2	Cluster 3
Successful dribbles per 90	2.003	0.223	1.254
Dribble Success (%)	56.983	68.761	55.137
Shot Assists per 90	1.560	0.163	0.816
Goal Assists per 90	0.182	0.017	0.080
Through Balls per 90	0.375	0.043	0.193
Crosses per 90	2.310	0.239	1.391
Carries per 90	41.408	5.402	25.460
Miscontrols per 90	2.175	0.366	1.690
Passes into box per 90	6.841	0.725	3.710

Table 4.3: Creativity cluster means

The cluster sizes were 46, 83, and 46 respectively. From the group means, we can deduce that each of the clusters identified represents a group of players with similar creative ability. It was possible that the clustering would identify certain groups of players that dribble a lot and do not cross, and another group that cross a lot but do not dribble much. This clustering has not identified groups like those, it has identified a group that excels in all features, a group that struggles in all

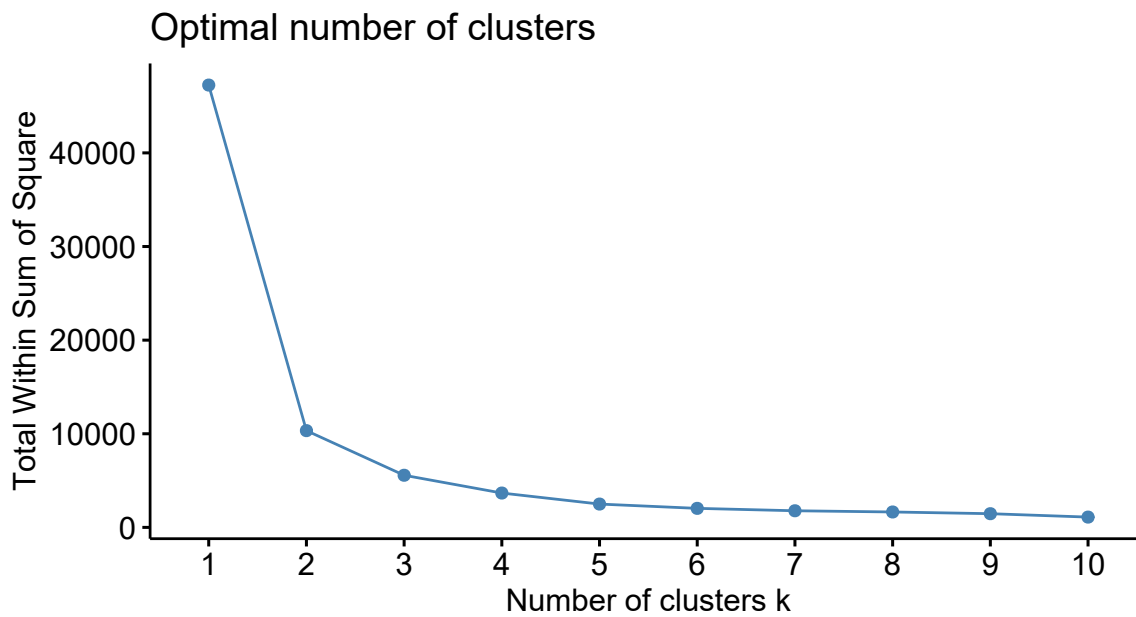


Figure 4.14: Sum of squared differences for creativity clustering

features, and one in between the two extremes. It is a hierarchal ordering in that sense. Cluster 1 contains the players with the best creative features, followed by cluster 3, then cluster 2 is at the bottom. The cluster means are highest in cluster 1 and lowest in cluster 2 for all of the features, other than the dribble success percentage, which we saw was independent to, or has weak negative correlation with, the other features. Interestingly, the cluster means for miscontrols is highest in cluster 1 and lowest in cluster 2 also. We might have expected players with superior creative ability to miscontrol the ball less, however, this clearly is not the case. This is probably due to players in cluster 1 getting on the ball more (highlighted by their superior number of carries per 90), which means that they have more opportunities to miscontrol the ball.

Here are some notable players in group 1: Adam Lallana, Alex Oxlade Chamberlain, Eden Hazard, Jack Grealish, Kevin De Bruyne, and Christian Eriksen. Notice that Lallana, Hazard, Grealish and Oxlade Chamberlain were all in cluster 1 for the passing clustering. We identified this cluster to contain players with lots of attacking responsibility, so we are happy to see these players in the top cluster for creativity, as expected. Furthermore, Mesut Ozil and David Silva, who were in cluster 2 for passing, are also in this top cluster for creativity. We predicted that they would be in

this cluster in that section, unlike most other players in that cluster that were not very attacking, and this confirms that.

The following players are in cluster 2: Alexander Mitrovic, Bastian Shweinsteiger, Charlie Adam, James Milner, James Morrison, Jermain Defoe, Mark Noble, Troy Deeney, and Wayne Rooney. The cluster means tell us that this cluster has the least creative output of the three. In here, we see some were in the top cluster for passing, such as Shweinsteiger, Milner, and Morrison. We are not surprised to see these here, as their passing statistics suggested that their job in their teams was to play deeper and make many passes, so they had little creative responsibility. We also see players like Rooney, Defoe, Mitrovic, and Deeney who generally play as strikers for their teams. Their job is to convert chances rather than create them. Cluster 3 contains players who were fairly creative, but not as much as cluster 1. Players in here include: Aaron Ramsey, Dele Alli, Fabio Borini, Heung Min Son, Jesse Lingard, Nacer Chadli, and Theo Wallcott.

Considering the two separate groups of players from cluster 3 in the passing t-SNE plot (those found at the bottom of the plot and those at the top), we see that all three examples given for players at the bottom of the plot (Milner, Adam, and Wanyama) were assigned to the bottom cluster for creativity. In contrast, the examples stated for the top section of the plot (Puncheon, Wijnaldum, and Stanislas) were each in the top or middle cluster for creativity. We theorized that their relative distance in the graph was due to the players at the top being more attacking. This clustering backs that up.

The t-SNE plot in figure 4.15 gives us a visualization of the players with respect to their creative performance. This plot accurately visualizes the clustering. If we drew a vertical line where Marc Albrighton is plotted on this graph (roughly the line $x = 15$) we see that all players to the right of this line are the players from cluster 1, the top performers. If we draw a horizontal line where Albrighton is plotted (roughly the line $y = -7$), we see that players below this, and to the left of the vertical line, are the players from cluster 2. The players in cluster 3 are found in the upper left section of this plot (with respect to these two theoretical lines).

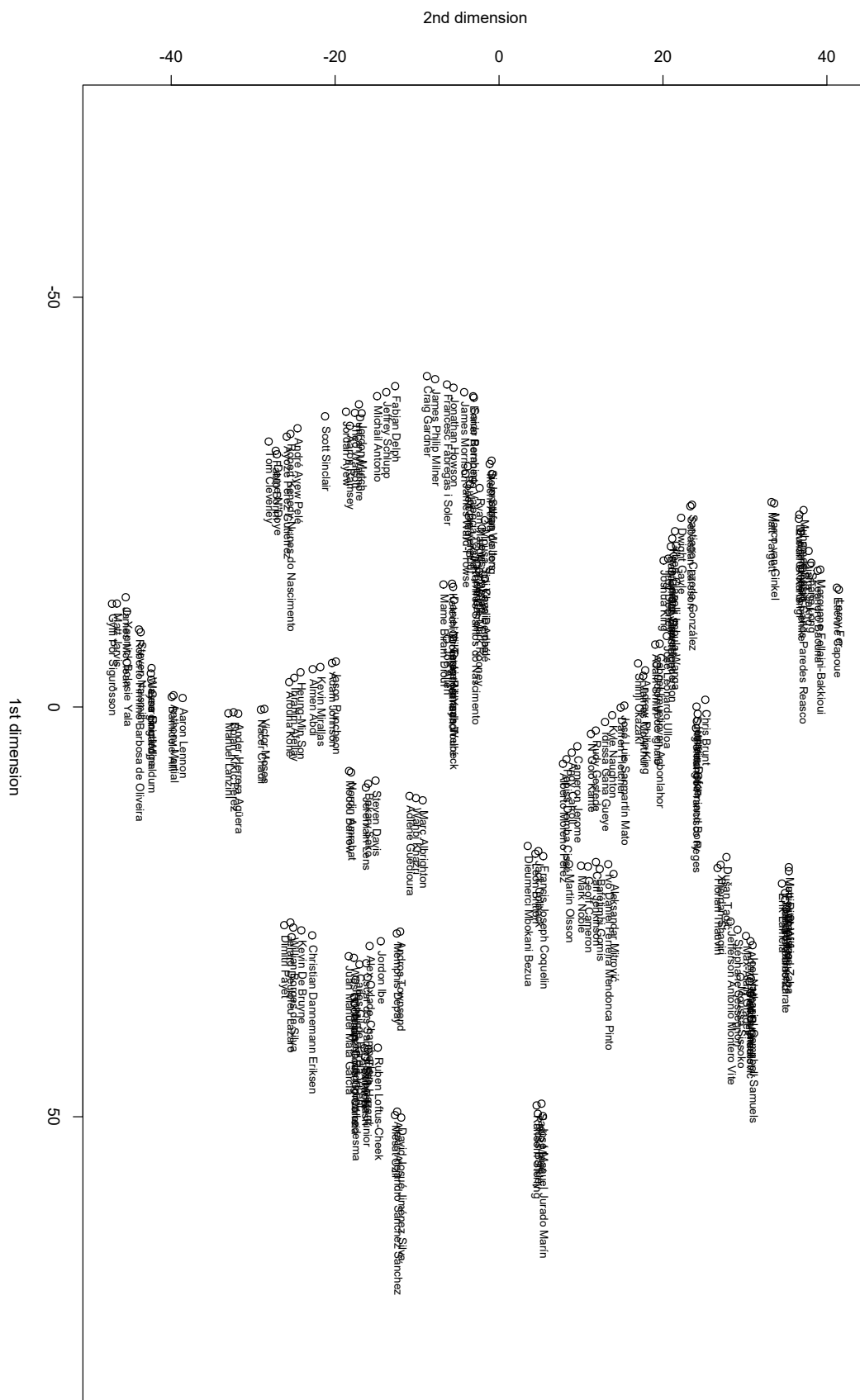


Figure 4.15: Creativity t-SNE reduction plot

4.1.4 Player comparison clustering summary and applications

In this section, we have seen three examples of k-means clustering being applied to player summary statistics. Specifically; finishing data, passing data, and creativity data. Considering the types of players assigned to each cluster in the three examples, we saw that all three examples made sense in a football context. This meant we were able to look deeper into the clusters and explore what insights we could learn from them.

The main insight we gained from all three examples was about the type of players in each cluster. In general, we were able to identify a rank ordering, where the 'best' players appeared in a certain cluster, the next best in another cluster, and so on. All three examples had a rank order of some degree, but the passing clustering was more complicated. The differences between players in some of the clusters were due to the roles of the players and not their ability in this example. We found that most of the creative players appeared in one cluster simply because they're play style means they don't make many passes. We also saw David Silva and Mesut Ozil are two exceptions to this.

Both of these players are very attacking. Most similar attacking players appeared in cluster 1, but because they often drop deeper in their teams and get involved in the build up play, they appeared instead in a cluster containing mostly center midfielders. When both of these appeared in the top cluster for creativity, this highlighted that these two had exceptional seasons in terms of both creativity and deeper passing build up play. From this we learn that comparing which clusters players are sorted into with respect to different data sets is a good way to get a better understanding of their performance for the season. We were also able to use the creativity clustering results to differentiate the different types of player in cluster 3 for passing. We conclude that using different data clustering together is an effective method to gain additional information.

In addition to identifying this rank order, we gained further insight by determining which features were responsible for the differences between clusters. In the finishing clustering, we learned that shooting accuracy is a key metric to indicate how successful a striker will be. In the passing clustering, we learned a more subtle fact that attacking midfielders make a higher percentage of short passes compared to defensive/central

midfielders.

We also gained some useful information from the accompanying t-SNE plots. These plots gave us a visual representation of the players, making it easier to see which players have similar statistics for each example. This was particularly useful for the passing example, as we learned more about cluster 3 here from the t-SNE plot. We saw that players in this cluster appeared in different areas of the t-SNE plot. Further analysis showed that cluster 3 had a mix of players similar to those in cluster 1 and others similar to those in cluster 2. These two groups of players were at the opposite ends of the t-SNE plot.

Before we move on, we should mention a possible application where player comparison clustering could be useful in football. The obvious area of football where this could be applied is in player recruitment. Clubs could collect and compile large data sets containing summary player statistics tailored to the metrics that they view as most important. I think the rank ordering we have found from player clustering could be a useful tool clubs could consider to help with their recruitment. Clubs could determine the rank ordering, then decide to only sign players from the top two ranked clusters, for example. By doing this, they have an easy way to rule out a large number of players immediately, which will save them time and allow them to consider a greater volume of players. Using t-SNE plots as an accompanying visual guide would also help clubs to see which groups of players they should consider signing and which they should exclude.

By applying clustering to larger, more complicated data sets for this purpose, clubs will likely also gain further insight about the game by analyzing the differences between clusters in a similar way to what we did for our three examples.

Chapter 5

Conclusion

During this project, we were successful in transforming the Statsbomb event data set into various more convenient forms, allowing us to apply clustering and t-SNE as we hoped. The aim was to create smaller datasets containing summary statistics about players' performances throughout the season (with respect to different areas of the game), and we were able to achieve this. The goal of the project was then to apply our methods to these data and see what we could learn. I felt this was an interesting topic to study as most current football data science uses regression models to model the game, so I was interested to see what results we could get from clustering methods. Specifically, I wanted to see how we could compare player performance in the season using k-means clustering and t-SNE as a visualization.

I decided to use k-means to perform the clustering as it is a simple, non-model-based clustering method ideal for exploratory data analysis. For our first example, I wanted to use a simplistic data set. This was done so that we had a simple example of the type of results we could obtain by clustering footballers. The hope was that the results made sense in a football context, meaning that it was a valid method we could use for more complex data sets.

For this purpose, I chose five simple finishing metrics as our first data set to apply clustering. Specifically, I chose shots per game, shots on target per game, goals per game, shooting accuracy(%) and shooting success(%). I was happy to see that the clustering made a lot of sense when I compared it to my prior knowledge of the players. All of the best strikers in the league, e.g. Marcus Rashford, Harry Kane, Sergio Agüero, and Jamie Vardy, were assigned to the same group. The clustering

identified three different levels of players with respect to finishing. The bottom group contained players who were particularly poor at finishing, whilst the players in either of the top two clusters were relatively proficient. Inspecting the group means, we discovered that the key difference between the players in either of the top two groups was their shooting accuracy. The players in the top group had significantly better shot accuracy than those in the middle group, resulting in significantly more goals being scored. This tells us that shooting accuracy is one of the most vital metrics that should be considered when signing a striker. Again, this makes sense in a football context, as a key coaching principle taught to all footballers is the importance of hitting the target.

This simple example showed us that k-means clustering can be used effectively to identify groups of players of similar ability. We also saw that we can gain insight about the game by identifying the features that were responsible for the clustering. We were very happy with these results and concluded that k-means clustering is an effective method to use for more complex football data sets.

The next example that we looked at contained 11 features that described the players' passing performance in the season. I restricted the data to midfielders, as I was most interested in midfielders' passing performance. The various features here included the number of passes per 90 minutes, the number of different length passes (short, medium, and long), the percentage of passes that were each length, and the passing success percentage for the different lengths of passes. I found three to be the optimal number of clusters, so I ran the clustering algorithm with $k = 3$. Looking at the cluster means, I noticed that the number of complete passes per 90 minutes was a key differentiator between the different types of player in each cluster. Cluster 1 only recorded 13.54 passes per 90 on average. This seemed very low, and by considering the players in this cluster, I realized that they were all very attacking, hence the lower number of passes. I also noticed that the players in this group had an elevated percentage of short passes. Studying this feature throughout the project, we learned that it was a good guide to understand how attacking a player was. The players in cluster 2 averaged 49.96 passes per 90 and those in cluster 3 averaged 32.16 passes per 90. We found that these two groups contained the deeper midfielders who control the game with their passing. It appeared that cluster 2 contained the best performers

in this position and cluster 3 contained less impressive players, this was highlighted by the higher number of passes for players in cluster 2 and their higher pass success percentage. We identified Mezut Ozil and David Silva as two outliers that we might have expected to be in cluster 1, as they are both very attacking minded, but they were in fact in cluster 2. This showed that they had exceptional seasons as they were involved in attacks, but also had similar passing statistics to the deeper midfielders.

The final example we looked at contained summary statistics about players' creative performance throughout the season. The data contained nine metrics that summarized the players' creativity, such as dribbles, assists, and crosses. I ran the clustering with $k = 3$, using the elbow method to determine this value for k . The result of this clustering was more simplistic than the passing clustering. It was more similar to the finishing example, with three levels of performance corresponding to the three clusters. The clustering made sense again in a footballing context, as all the top creative players in the league were indeed assigned to the top cluster (De Bruyne, Hazard, Ozil). Most of the players who were in the top group for this example were in cluster 1 for the passing example. We expected this as their attacking responsibility meant they generally don't get involved in the deeper passing build up play. Ozil and Silva were both in the top cluster here, confirming that they are anomalies in cluster 2 for passing. Indeed, most of the other players in cluster 2 for passing were in the bottom cluster here, as we explained, they were generally center midfielders who did not have much creative responsibility in their teams.

During the three examples, we learned that k-means clustering can be used to find groups of players with similar abilities in football, based on specific areas of the game. This could be a useful method for top of the funnel scouting, as clubs can identify players in the bottom clusters, and class them as unsuitable options to bring to their team. They could tailor this approach so that the data contains all of the features that they view as important. This could save clubs large amounts of time scouting if the list of possible players is very large. It will also allow them to consider far more players than they would have otherwise, as they have an easy way to rule out a large proportion of them. Given this highly useful possible application, I believe there is plenty of scope for further research on this topic. We could use more advanced clustering methods and more complex data to obtain more powerful

insights.

We also saw in two of these examples, finishing and passing, that we can learn insights about the game itself by studying the different cluster means. In the finishing example, we deduced a trivial fact; hitting the target is a vital attribute for top strikers. The passing example gave us a more interesting insight, as we learned that attacking midfielders have relatively high percentages of short passes. These suggest that clustering footballers using specific data sets can teach us about the game itself, in addition to what we learn about the players.

We conclude that player comparison clustering is an effective method for football data analysis that should be explored further. With more resources, clustering could be performed on larger data sets with a high number of features. There are many more areas of football where we could apply this, such as defensive performance or goalkeeping. There are opportunities to learn a lot about players and the game using these methods.

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Appendix A

Computer code

For reproducibility, here is the computer code that I have used.

For code, visit:

<https://github.com/Furseybutler/Football-Data-Analysis-Project>

For more information about the data, visit:

<https://github.com/statsbomb/open-data/blob/master/doc>